

1: Floyd's

	1	2	3	4	5	
1	0	1	∞	1	5	1
2	9	0	3	2	∞	2
3	∞	∞	0	4	∞	3
4	∞	∞	2	0	3	4
5	3	∞	∞	∞	0	5

Step 1:

$$(2,1) = 9 \quad (1,2) = 1 \quad (2,2) = 0$$

$$(2,2) = \min((2,2), (2,1) + (1,2)) = (0,10) = 0$$

$$(2,1) = 9 \quad (1,4) = 1 \quad (2,4) = (2,10) = 2$$

$$(2,1) = 9 \quad (1,5) = 5 \quad (2,5) = (\infty,14) = 14$$

$$(5,1) = 3 \quad (1,2) = 1 \quad (5,2) = (\infty,4) = 4$$

$$(5,1) = 3 \quad (1,4) = 1 \quad (5,4) = (\infty,4) = 4$$

$$(5,1) = 3 \quad (1,5) = 5 \quad (5,5) = (0,\infty) = 0$$

	1	2	3	4	5	
1	0	1	∞	1	5	1
2	9	0	3	2	14	2
3	∞	∞	0	4	∞	3
4	∞	∞	2	0	3	4
5	3	4	∞	4	0	5

Step 2 : $(1,2) = 1 \quad (2,1) = 9 \quad (1,1) = (0,10) = 0$

$$(1,2) = 1 \quad (2,3) = 3 \quad (1,3) = (\infty,4) = 4$$

$$(1,2) = 1 \quad (2,4) = 2 \quad (1,4) = (1,3) = 1$$

$$(1,2) = 1 \quad (2,5) = 14 \quad (1,5) = (5,15) = 5$$

$$(5,2) = 4 \quad (2,1) = 9 \quad (5,1) = (3,13) = 3$$

$$(5,2) = 4 \quad (2,3) = 3 \quad (5,3) = (\infty,7) = 7$$

$$(5,2) = 4 \quad (2,4) = 2 \quad (5,4) = (4,6) = 4$$

$$(5,2) = 4 \quad (2,5) = 14 \quad (5,5) = (0,18) = 0$$

Step 3:

	1	2	3	4	5	
1	0	1	4	1	5	1
2	9	0	3	2	14	2
3	∞	∞	0	4	∞	3
4	∞	∞	2	0	3	4
5	3	4	7	4	0	5

$$(1,3) = 4 \quad (3,4) = 4 \quad (1,4) = (1,8) = 1$$

$$(2,3) = 3 \quad (3,4) = 4 \quad (2,4) = (2,7) = 2$$

$$(4,3) = 2 \quad (3,4) = 4 \quad (4,4) = (0,6) = 0$$

$$(5,3) = 7 \quad (3,4) = 4 \quad (5,4) = (4,11) = 4$$

Step 4:

	1	2	3	4	5	
1	0	1	4	1	5	1
2	9	0	3	2	14	2
3	∞	∞	0	4	∞	3
4	∞	∞	2	0	3	4
5	3	4	7	4	0	5

$$(1,4) = 1 \quad (4,3) = 2 \quad (1,3) = (4,3) = 3$$

$$(1,4) = 1 \quad (4,5) = 3 \quad (1,5) = (5,4) = 4$$

$$(2,4) = 2 \quad (4,3) = 2 \quad (2,3) = (3,4) = 3$$

$$(2,4) = 2 \quad (4,5) = 3 \quad (2,5) = (14,5) = 5$$

$$(3,4) = 4 \quad (4,3) = 2 \quad (3,3) = (0,6) = 0$$

$$(3,4) = 4 \quad (4,5) = 3 \quad (3,5) = (\infty,7) = 7$$

$$(5,4) = 4 \quad (4,3) = 2 \quad (5,3) = (7,6) = 6$$

$$(5,4) = 4 \quad (4,5) = 3 \quad (5,5) = (0,7) = 0$$

Step 5:

	1	2	3	4	5	
1	0	1	4	1	4	1
2	9	0	3	2	5	2
3	∞	∞	0	4	7	3
4	∞	∞	2	0	3	4
5	3	4	6	4	0	5

$$(1,5) = 4 \quad (5,1) = 3 \quad (1,1) = (0,7) = 0$$

$$(1,5) = 4 \quad (5,2) = 4 \quad (1,2) = (1,8) = 1$$

$$(1,5) = 4 \quad (5,3) = 6 \quad (1,3) = (3,10) = 3$$

$$(1,5) = 4 \quad (5,4) = 4 \quad (1,4) = (1,8) = 1$$

$$(2,5) = 5 \quad (5,1) = 3 \quad (2,1) = (9,8) = 8$$

$$(2,5) = 5 \quad (5,2) = 4 \quad (2,2) = (0,9) = 0$$

$$(2,5) = 5 \quad (5,3) = 6 \quad (2,3) = (3,11) = 3$$

$$(2,5) = 5 \quad (5,4) = 4 \quad (2,4) = (2,9) = 2$$

$$(3,5) = 7 \quad (5,1) = 3 \quad (3,1) = (\infty,10) = 10$$

$$(3,5) = 7 \quad (5,2) = 4 \quad (3,2) = (\infty,11) = 11$$

$$(3,5) = 7 \quad (5,3) = 6 \quad (3,3) = (0,10) = 10$$

$$(3,5) = 7 \quad (5,4) = 4 \quad (3,4) = (4,11) = 4$$

$$(4,5) = 3 \quad (5,1) = 3 \quad (4,1) = (\infty,6) = 6$$

$$(4,5) = 3 \quad (5,2) = 4 \quad (4,2) = (\infty,7) = 7$$

$$(4,5) = 3 \quad (5,3) = 6 \quad (4,3) = (2,9) = 2$$

$$(4,5) = 3 \quad (5,4) = 4 \quad (4,4) = (0,7) = 0$$

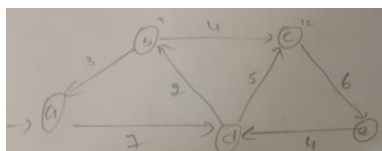
0	1	4	1	4
8	0	3	2	5
10	11	0	4	7
6	7	2	0	3
3	4	6	4	0

Item	Weight	Value
1	2	12
2	1	10
3	3	20
4	2	15

	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	12	12	12	12	12
2	0	12	12	22	22	22
3	0	22	22	22	22	32
4	0	32	32	32	32	37

i	j	V[i, j]	$V[i,j] = \text{Max}(v[i-1,j], v[i-1,j-w[i]]+p[i])$
4	5	V[4,5]	$V[4,5] = \text{Max}(v[3,5], v[3,3]+15) = 37$
3	5	V[3,5]	$V[3,5] = \text{Max}(v[2,5], v[2,2]+20) = 32$
3	3	V[3,3]	$V[3,3] = \text{Max}(v[2,3], v[2,0]+20) = 22$
2	5	V[2,5]	$V[2,5] = \text{Max}(v[1,5], v[1,4]+10) = 22$
2	2	V[2,2]	$V[2,2] = \text{Max}(v[1,2], v[1,1]+10) = 12$
1	5	V[1,5]	$V[1,5] = \text{Max}(v[0,5], v[0,3]+12) = 12$
1	4	V[1,4]	$V[1,4] = \text{Max}(v[0,4], v[0,2]+12) = 12$
1	2	V[1,2]	$V[1,2] = \text{Max}(v[0,2], v[0,0]+12) = 12$
1	1	V[1,1]	$V[1,1] = 0$
2	3	V[2,3]	$V[2,3] = \text{Max}(v[1,3], v[1,2]+10) = 22$

3. Single Source Shortest Path(DJ)



a b c d e

0	∞	∞	7	∞
3	0	4	∞	∞
∞	∞	0	∞	6
∞	2	5	0	∞
∞	∞	∞	4	0

a
b
c
d
e

d[a]	0	0
d[b]	∞	9
d[c]	∞	12
d[d]	7	7
d[e]	∞	∞

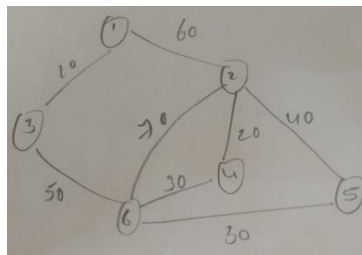
s	V=V-s	$d[v] = \min(d[v], d[u] + \text{cost}[u][v])$	$v \in V-s$
a	-	---	d
a,d	b,c,e	$d[b] = \min(d[b], d[d] + \text{cost}[d,b]) = (\infty, 7+2) = 9$	b

		$d[c] = \min(d[c], d[d] + \text{cost}[d,c]) = (\infty, 7+5) = 12$ $d[e] = \min(d[e], d[d] + \text{cost}[d,e]) = (\infty, 7+\infty) = \infty$	
a,d,b	c,e	$d[c] = \min(d[c], d[b] + \text{cost}[b,c]) = (12, 9+4) = 12$ $d[e] = \min(d[e], d[b] + \text{cost}[b,e]) = (\infty, 9+\infty) = \infty$	c
a,d,b,c	e	$d[e] = \min(d[e], d[c] + \text{cost}[c,e]) = (\infty, 12+6) = 18$	e

d[a] d[b] d[c] d[d] d[e]

0	9	12	7	18
---	---	----	---	----

4 . Kruskal's



1-3 10

2-4 20

4-6 30

5-6 30

2-5 40

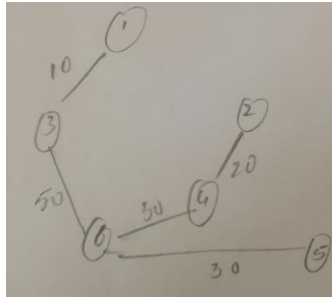
3-6 50

1-2 60

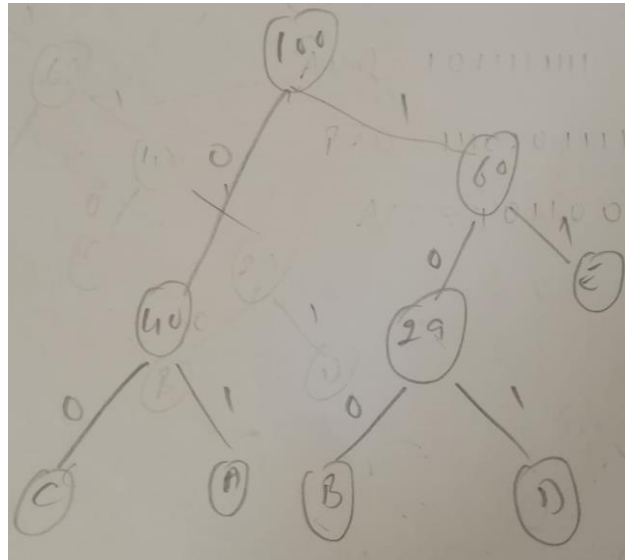
2-6 70

Steps	Deleted Edge (u,v)	i=find(u) j=find(v)	o/p (u,v)	Union(i,j) S[1],s[2],s[3],s[4],s[5],s[6]
Initial	-	-	-	1 2 3 4 5 6
1	1-3	1,3	1-3	1 2 1 4 5 6
2	2-4	2,4	2-4	1 2 1 2 5 6
3	4-6	2,6	4-6	1 2 1 2 5 2
4	5-6	5,2	5-6	1 2 1 2 2 2
5	2-5	2,2	Discard	1 2 1 2 2 2
6	3-6	1,2	3-6	1 1 1 2 2 2
7	1-2	1,1	Discard	1 1 1 2 2 2
8	2-6	1,1	Discard	1 1 1 2 2 2

$$10+50+30+30+20 = 140$$



5. A: 22 B: 13 C: 18 D: 16 E: 31



C A B = 0001100

ADD = 01101101

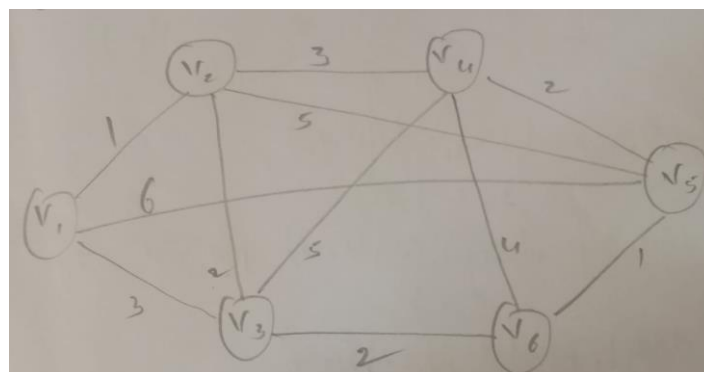
BAD = 10001101

ACE = 010011

110011 = ECE

1000110001 = BABA

6. Prim's



v1 v2 v3 v4 v5 v6

0	1	3	∞	6	∞	v1
1	0	2	3	5	∞	v2
3	2	0	5	∞	2	v3
∞	3	5	0	2	4	v4
6	5	∞	2	0	1	v5
∞	∞	2	4	1	0	v6

d[v1] d[v2] d[v3] d[v4] d[v5] d[v6]

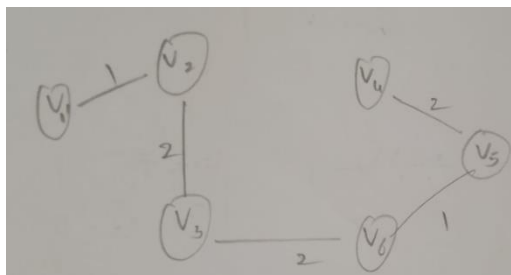
0	1	3	∞	6	∞
---	---	---	----------	---	----------

p[v1] p[v2] p[v3] p[v4] p[v5] p[v6]

0	V1	V2	V5	V6	V3
---	----	----	----	----	----

s	v=V-s	$d[v]=\min(d[v], \text{cost}[u][v])$	P[v]=u	Find u,d[u]	o/p u,p[u]
v1	v2,v3,v4,v5,v6	--	-	v2,1	v2,v1
v1,v2	v3,v4,v5,v6	$d[v3]=\min(d[v3], \text{cost}[v2,v3])=(3,2)=2$ $d[v4]=\min(d[v4], \text{cost}[v2,v4])=(\infty,3)=3$ $d[v5]=\min(d[v5], \text{cost}[v2,v5])=(6,5)=5$ $d[v6]=\min(d[v6], \text{cost}[v2,v6])=(\infty,\infty)=\infty$	p[v3]=v2	v3,2	v3,v2
v1,v2,v3	v4,v5,v6	$d[v4]=\min(d[v4], \text{cost}[v3,v4])=(\infty,5)=5$ $d[v5]=\min(d[v5], \text{cost}[v3,v5])=(6,\infty)=6$ $d[v6]=\min(d[v6], \text{cost}[v3,v6])=(\infty,2)=2$	p[v6]=v3	v6,2	v6,v3
v1,v2,v3,v6	v4,v5	$d[v4]=\min(d[v4], \text{cost}[v6,v4])=(\infty,4)=4$ $d[v5]=\min(d[v5], \text{cost}[v6,v5])=(6,1)=1$	p[v5]=v6	v5,1	v5,v6
v1,v2,v3,v6,v5	v4	$d[v4]=\min(d[v4], \text{cost}[v5,v4])=2$	p[v4]		

$$1+2+2+2+1=8$$



Using Branch and Bound technique solve the below instance of knapsack problem.

Item	Weight	Value
1	2	12
2	1	10
3	3	20
4	2	5

Sol:

Item	Weight	Value	V_{i+1}/W_{i+1}
1	1	10	10 v_1/w_1
2	2	12	6 v_2/w_2
3	3	20	6 v_3/w_3
4	2	5	2 v_4/w_4

$$M=5$$

Step: 1 $i=0$ $w=0$ $v=0$ $v_1/w_1=10$

$$M-w = 5-0 = 5$$

$$ub = 0+5*10 = 50$$

Step: 2 $I = 1$ $v_2/w_2 = 6$

With item 1

$$W = \text{weight item 1} + w = 1+0 = 1$$

$$V = v+v_1 = 0+10 = 10$$

$$M-w = 5-1 = 4$$

$$ub = 10+4*6=34$$

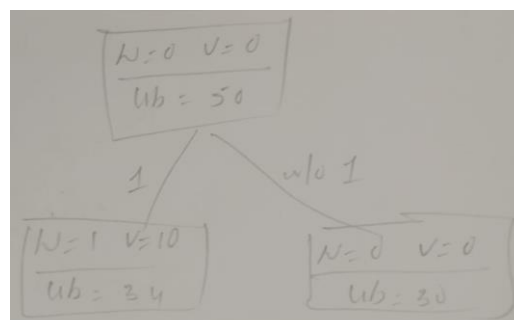
Without Item 1

$$w = 0+0=0$$

$$v = 0+0 = 0$$

$$M-w=5-0=5$$

$$ub=0+5*6= 30$$



Step 3

Item 2 $v_3/w_3 = 6$

With item 2

$$W = \text{weight item 2} + w = 2+1 = 3$$

$$V = v+21 = 10+12 = 22$$

$$M-w = 5-3 = 2$$

$$ub = 22+2*6=34$$

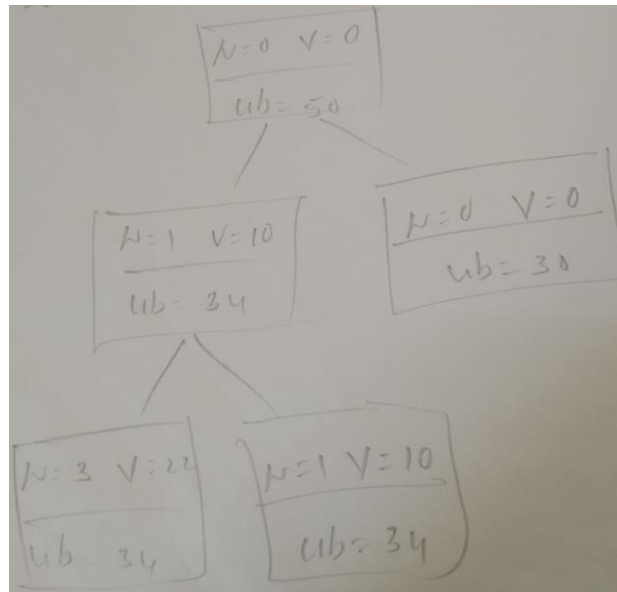
Without Item 2

$$w = 0+1=1$$

$$v = 10+0 = 10$$

$$M-w=5-1=4$$

$$ub=10+6*4= 34$$



Step 4

Item 3 $v_4/w_4 = 2$

With item 3

$$W = \text{weight item 2} + w = 3+3 = 6$$

Not Feasible

Without Item 3

$$w = 0+3=3$$

$$v = 0+22 = 22$$

$$M-w=5-3=2$$

$$ub=22+2*2= 26$$

Step 5

Item 4 $v_5/w_5 = ?$

With item 4

$$W = \text{weight item 4} + w = 2+3 = 5$$

$$V = v+21 = 22+5 = 27$$

$$ub = v \quad ub = 27$$

Without Item 4

$$w = 0+w=0+3=1$$

$$v = 22+0 = 22$$

$$ub=v \quad ub=22$$

